

Table 12: Detailed LLM Performances on COGS.

Model	Direct	CoT		ReAct		ToT	
	Acc (%)	Acc (%)	# tokens	Acc (%)	# tokens	Acc (%)	# tokens
<i>Open-source</i>							
Deepseek-V3	30.80	14.00 (-16.80)	405.03	27.80 (-3.00)	375.88	15.10 (-15.70)	729.94
Gemma2-9B	9.10	8.50 (-0.60)	309.71	6.70 (-2.40)	223.18	1.30 (-7.80)	328.61
Gemma2-27B	21.30	14.10 (-7.20)	132.83	11.40 (-9.90)	243.44	7.90 (-13.40)	267.56
Llama3.1-8B	8.10	3.60 (-4.50)	415.50	2.80 (-5.30)	628.31	2.40 (-5.70)	1072.55
Llama3.1-70B	20.80	17.40 (-3.40)	207.46	12.20 (-8.60)	348.43	4.70 (-16.10)	669.70
Llama3.1-405B	24.40	13.80 (-10.60)	304.01	8.20 (-16.20)	395.69	3.20 (-21.20)	693.21
Qwen2.5-7B	11.50	9.90 (-1.60)	241.29	7.10 (-4.40)	310.74	5.00 (-6.50)	511.24
Qwen2.5-72B	20.40	19.20 (-1.20)	196.61	12.40 (-8.00)	86.36	17.50 (-2.90)	593.44
Mistral-7B	7.60	5.20 (-2.40)	152.42	2.70 (-4.90)	139.00	3.10 (-4.50)	163.96
Mistral-Small 3	15.40	12.70 (-2.70)	369.97	12.90 (-2.50)	192.04	8.30 (-7.10)	595.81
<i>Proprietary</i>							
Gemini-1.5-flash	23.70	19.20 (-4.50)	181.79	11.50 (-12.20)	293.78	4.95 (-18.75)	459.69
Gemini-1.5-pro	36.00	28.10 (-7.90)	226.60	25.29 (-10.71)	344.10	22.50 (-13.50)	463.41
Gemini-2.0-flash	31.60	28.70 (-2.90)	231.77	25.20 (-6.40)	216.32	14.81 (-16.79)	425.68
GPT-3.5-turbo	11.60	8.40 (-3.20)	113.25	8.50 (-3.10)	116.25	7.89 (-3.71)	138.12
GPT-4o-mini	18.10	13.50 (-4.60)	171.29	11.90 (-6.20)	233.30	10.10 (-8.00)	247.97
GPT-4o	25.30	21.70 (-3.60)	246.19	21.30 (-4.00)	208.00	19.16 (-6.14)	397.85
Average	19.73	14.88 (-4.85)	244.11	12.99 (-6.74)	272.18	9.24 (-10.49)	484.92

Table 13: Detailed LLM Performances on SALT.

Model	Direct	CoT		ReAct		ToT	
	Acc (%)	Acc (%)	# tokens	Acc (%)	# tokens	Acc (%)	# tokens
<i>Open-source</i>							
Deepseek-V3	52.83	45.83 (-7.00)	161.16	36.08 (-16.75)	419.34	41.92 (-10.91)	549.59
Gemma2-9B	13.83	19.17 (+5.34)	139.96	17.08 (+3.25)	270.40	15.33 (+1.50)	294.55
Gemma2-27B	39.33	34.33 (-5.00)	100.25	26.67 (-12.66)	222.94	11.17 (-28.16)	323.54
Llama3.1-8B	25.25	17.92 (-7.33)	253.14	15.17 (-10.08)	505.72	15.00 (-10.25)	1082.77
Llama3.1-70B	48.33	44.92 (-3.41)	204.69	41.42 (-6.91)	339.29	37.67 (-10.66)	602.01
Llama3.1-405B	57.92	59.67 (+1.75)	430.02	54.17 (-3.75)	435.96	33.50 (-24.42)	616.12
Qwen2.5-7B	15.58	13.33 (-2.25)	116.37	11.00 (-4.58)	165.47	10.83 (-4.75)	318.66
Qwen2.5-72B	35.25	39.42 (+4.17)	137.75	38.33 (+3.08)	240.15	43.50 (+8.25)	324.49
Mistral-7B	14.92	11.75 (-3.17)	112.82	11.17 (-3.75)	148.71	5.42 (-9.50)	357.95
Mistral-Small 3	44.67	36.25 (-8.42)	167.56	33.17 (-11.50)	274.82	28.67 (-16.00)	584.81
<i>Proprietary</i>							
Gemini-1.5-flash	39.50	38.08 (-1.42)	201.34	41.58 (+2.08)	456.13	34.42 (-5.08)	542.69
Gemini-1.5-pro	53.50	45.92 (-7.58)	213.63	43.83 (-9.67)	476.21	42.92 (-10.58)	387.65
Gemini-2.0-flash	53.67	48.92 (-4.75)	143.62	51.08 (-2.59)	403.63	47.50 (-6.17)	762.87
GPT-3.5-turbo	26.50	20.33 (-6.17)	107.55	15.50 (-11.00)	149.05	15.00 (-11.50)	154.97
GPT-4o-mini	31.83	24.17 (-7.66)	151.52	18.08 (-13.75)	245.74	11.67 (-20.16)	540.46
GPT-4o	50.67	46.41 (-4.26)	174.41	42.67 (-8.00)	309.02	41.42 (-9.25)	440.49
Average	37.72	34.15 (-3.57)	175.99	31.06 (-6.66)	316.41	27.25 (-10.47)	492.73

Table 14: Detailed LLM Performances on List Function.

Model	Direct	CoT		ReAct		ToT	
	Acc (%)	Acc (%)	# tokens	Acc (%)	# tokens	Acc (%)	# tokens
<i>Open-source</i>							
Deepseek-V3	54.88	58.32 (+3.44)	487.50	34.56 (-20.32)	457.90	47.20 (-7.68)	476.20
Gemma2-9B	32.80	25.20 (-7.60)	146.12	22.24 (-10.56)	193.28	21.04 (-11.76)	370.36
Gemma2-27B	43.60	35.44 (-8.16)	98.08	34.88 (-8.72)	186.05	36.40 (-7.20)	243.89
Llama3.1-8B	35.20	24.48 (-10.72)	435.90	18.48 (-16.72)	621.80	15.12 (-20.08)	978.29
Llama3.1-70B	41.28	37.92 (-3.36)	171.67	38.16 (-3.12)	312.06	38.40 (-2.88)	578.48
Llama3.1-405B	50.40	47.76 (-2.64)	425.44	47.68 (-2.72)	410.83	19.68 (-30.72)	545.26
Qwen2.5-7B	37.60	29.36 (-8.24)	382.33	21.36 (-16.24)	144.06	24.64 (-12.96)	348.04
Qwen2.5-72B	49.28	45.04 (-4.24)	427.86	42.96 (-6.32)	266.22	41.92 (-7.36)	380.14
Mistral-7B	28.00	8.96 (-19.04)	161.29	4.96 (-23.04)	129.42	2.00 (-26.00)	392.23
Mistral-Small 3	41.76	38.32 (-3.44)	510.16	36.00 (-5.76)	224.67	32.32 (-9.44)	524.35
<i>Proprietary</i>							
Gemini-1.5-flash	46.96	42.00 (-4.96)	369.28	42.72 (-4.24)	485.17	39.44 (-7.52)	567.06
Gemini-1.5-pro	53.28	52.00 (-1.28)	341.34	53.60 (+0.32)	447.85	46.32 (-6.96)	456.78
Gemini-2.0-flash	53.76	51.12 (-2.64)	332.87	53.04 (-0.72)	451.64	40.56 (-13.20)	895.93
GPT-3.5-turbo	42.16	31.12 (-11.04)	131.08	25.92 (-16.24)	99.82	24.48 (-17.68)	154.43
GPT-4o-mini	44.88	36.88 (-8.00)	188.62	35.04 (-9.84)	228.55	34.56 (-10.32)	331.80
GPT-4o	53.04	48.72 (-4.32)	211.48	45.84 (-7.20)	311.21	35.92 (-17.12)	444.89
Average	44.58	38.80 (-5.78)	305.49	35.25 (-9.33)	310.73	31.63 (-12.95)	486.49

Table 15: Detailed LLM Performances on RAVEN.

Model	Direct	CoT		ReAct		ToT	
	Acc (%)	Acc (%)	# tokens	Acc (%)	# tokens	Acc (%)	# tokens
<i>Open-source</i>							
Deepseek-V3	21.05	8.98 (-12.07)	397.14	6.27 (-14.78)	413.69	2.14 (-18.91)	780.90
Gemma2-9B	13.74	1.99 (-11.75)	252.07	0.87 (-12.87)	298.49	0.08 (-13.66)	435.18
Gemma2-27B	16.60	1.11 (-15.49)	196.51	1.91 (-14.69)	287.23	0.24 (-16.36)	349.96
Llama3.1-8B	7.07	0.87 (-6.20)	456.49	1.43 (-5.64)	757.91	0.32 (-6.75)	943.80
Llama3.1-70B	17.08	9.93 (-7.15)	325.79	5.24 (-11.84)	576.85	1.19 (-15.89)	1022.81
Llama3.1-405B	25.34	16.92 (-8.42)	355.79	11.68 (-13.66)	543.75	3.97 (-21.37)	843.70
Qwen2.5-7B	11.12	0.56 (-10.56)	673.17	1.59 (-9.53)	545.11	0.24 (-10.88)	964.45
Qwen2.5-72B	23.67	8.18 (-15.49)	476.07	6.35 (-17.32)	588.40	4.92 (-18.75)	608.39
Mistral-7B	1.51	0.00 (-1.51)	296.05	0.08 (-1.43)	326.30	0.16 (-1.35)	512.38
Mistral-Small 3	16.44	10.41 (-6.03)	828.07	5.80 (-10.64)	475.78	2.14 (-14.30)	841.24
<i>Proprietary</i>							
Gemini-1.5-flash	19.06	5.88 (-13.18)	478.57	3.02 (-16.04)	644.90	1.27 (-17.79)	666.44
Gemini-1.5-pro	24.31	11.12 (-13.19)	545.27	12.87 (-11.44)	728.48	7.94 (-16.37)	564.41
Gemini-2.0-flash	23.35	19.78 (-3.57)	719.08	20.02 (-3.33)	1017.31	13.66 (-9.69)	1634.67
GPT-3.5-turbo	12.79	5.80 (-6.99)	206.14	3.57 (-9.22)	131.61	3.73 (-9.06)	235.38
GPT-4o-mini	15.65	6.12 (-9.53)	322.94	3.18 (-12.47)	337.58	0.95 (-14.70)	598.97
GPT-4o	22.24	10.33 (-11.91)	427.07	8.90 (-13.34)	469.12	6.43 (-15.81)	718.92
Average	17.25	7.37 (-9.88)	434.75	5.87 (-11.38)	533.09	2.84 (-14.41)	737.64